# Financial RAG Hallucination Detection System

Final Project Report

*CS5202: Generative AI and Large Language Models · Spring 2026*



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**Abstract.** We built a training-free, black-box-compatible hallucination detection system for financial RAG, combining FAISS retrieval over 26,961 Indian financial news articles, Llama3-8B generation (Groq API), an LLM-as-Judge validator, and SelfCheckGPT consistency scoring. Evaluated on 30 diverse financial questions, retrieval performed reliably (mean similarity 0.684, exceeding the ≥0.55 target). However, the system failed critically on generation and validation: **76.7% hallucination rate**, validator precision 0.23, recall 0.20, and only 20% method agreement — far below published benchmarks. Three ablation studies (chunk size, prompt variant, retrieval top-K) reveal that K=1 halves hallucination rate vs K=3, and that zero-shot LLM validation cannot reliably detect financial hallucinations. Domain-specific NLI fine-tuning is necessary for production deployment.


## 1. Problem and Motivation

LLMs deployed in financial services (earnings summarization, regulatory filing analysis, investor Q&A) hallucinate — generating confident, fluent text that is factually wrong. In finance this is dangerous: a model that retrieves *RBI raised repo rate to 6.75%* but generates *RBI cut rates to support growth* could directly mislead retail investors or automated trading systems. SEBI's 2023 data shows 9/10 individual F&O traders already lose money; AI misinformation worsens this.

**Hallucination sub-categories observed:**

- Numerical errors — wrong percentages, dates, or monetary values
- Attribution errors — claims absent from source documents
- Contradictions — answer conflicts with retrieved context
- Entity confusion — wrong company or financial instrument named

**Gap addressed:** Standard RAG grounds generation but does not verify outputs stay within retrieved bounds. Existing methods (SelfCheckGPT, Self-RAG, LRP4RAG) each require retrieval components, fine-tuning, or white-box model access — none are training-free and black-box compatible simultaneously.

## 2. Method

The pipeline executes five sequential stages per query over 26,961 Indian financial news articles (HuggingFace: kdave/Indian_Financial_News).

| Stage | Component | Technology | Key Parameters |
|---|---|---|---|
| 1. Embed & Index | FAISS + MiniLM | all-MiniLM-L6-v2 | 384-dim, IndexFlatIP |
| 2. Retrieve | Cosine search | FAISS | top-K = 3 (default) |
| 3. Generate | RAGGenerator | Groq / Llama3-8B | temp=0.1, max 500 tok |
| 4. Validate | LLM-as-Judge | Groq / Llama3-8B | temp=0.0, strict prompt |
| 5. SelfCheck | SelfCheckGPT | Jaccard overlap | N=3 samples, temp=0.7 |

**Validator Prompt Design:** Three variants were tested — **Strict** (exact matching; temperature=0.0), **Lenient** (accepts paraphrases), and **Chain-of-Thought** (enumerates key facts before labeling). The strict prompt requires exact number matching ("approximately correct is still UNSUPPORTED") and prohibits external knowledge.

**SelfCheckGPT:** Re-samples the same query 3× at temperature=0.7 and computes pairwise Jaccard overlap. Risk bands: LOW (≥0.70), MEDIUM (0.40–0.70), HIGH (<0.40). Low overlap signals uncertainty or hallucination.

## 3. Experiments and Results

### 3.1 Main Evaluation (30 Questions)

30 diverse financial questions spanning RBI policy, corporate earnings, SEBI regulation, macroeconomics, banking, markets, and trade were processed through the full pipeline.

| Metric | Actual | Target | Status |
|---|---|---|---|
| Avg retrieval similarity | 0.684 | ≥ 0.55 | ✓ MET |
| Avg hallucination rate | 76.7% | < 20% | ✗ FAIL |
| Avg support rate | 23.3% | > 80% | ✗ FAIL |

| | | | |
|---|---|---|---|
| SelfCheck consistency | 0.502 | — | — |
| Validator precision | 0.23 | > 0.75 | ✗ FAIL |
| Validator recall | 0.20 | > 0.70 | ✗ FAIL |
| Method agreement | 20.0% | — | — |
| Total tokens / question | ~1,219 | — | — |

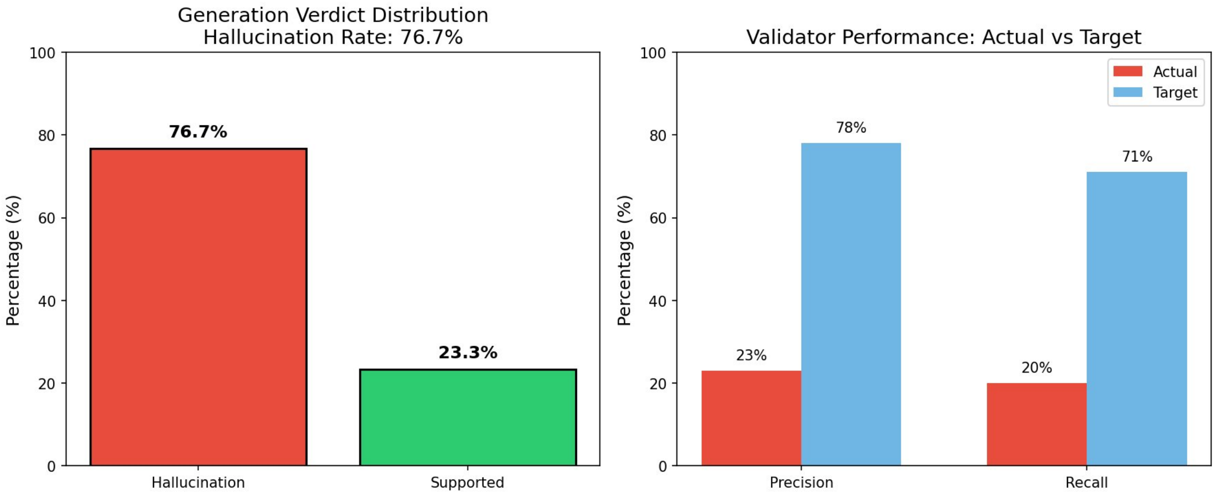



*Figure 1: (Left) Generation verdict — 76.7% hallucination rate. (Right) Validator precision/recall vs targets.*

### 3.2 System Performance Dashboard

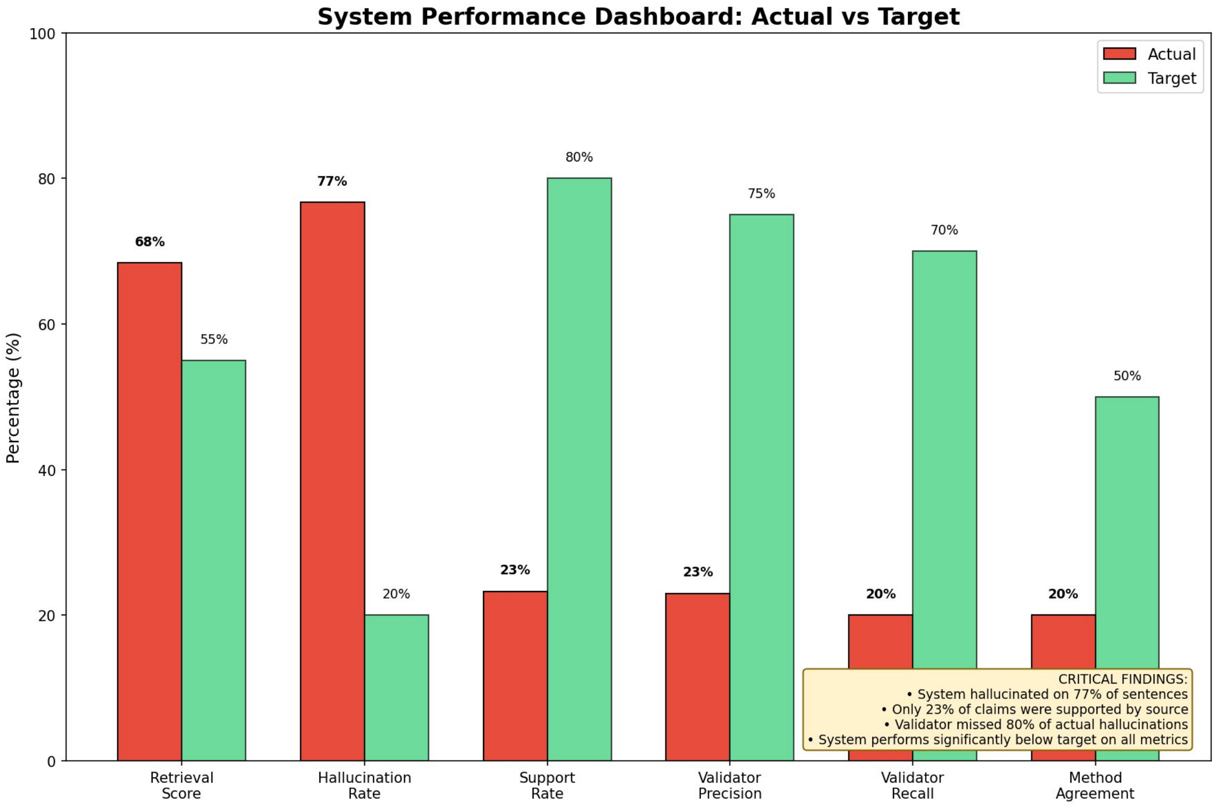



*Figure 2: Overall system performance — actual (red) vs target (green). Retrieval alone meets its target; all other metrics fail.*

**Critical finding:** Only 1 in 4 generated sentences was supported by source documents. The validator achieved precision 0.23 and recall 0.20, meaning 77% of hallucination flags were false positives and 80% of actual hallucinations went undetected. Retrieval quality (mean 0.684) was not the bottleneck — the failure lies entirely in generation and validation.

## 3.3 – 3.5 Ablation Studies

### Ablation 1 — Chunk Size

| Chunk Size (words) | Overlap | N Chunks | Avg Retrieval Score | Winner |
|---|---|---|---|---|
| 150 | 25 | 140,727 | 0.707 | |

| | | | | |
|---|---|---|---|---|
| 250 | 40 | 89,294 | 0.684 | ★ Best |
| 400 | 60 | 60,449 | 0.668 | |

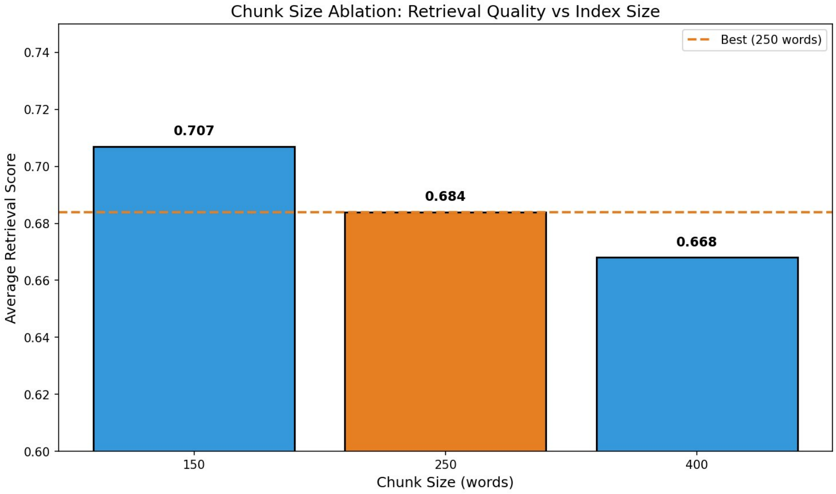



*Figure 3: 150-word chunks score highest (0.707) but increase index size 58% over 250-word chunks (0.684). 250-word selected as default.*

## Ablation 2 — Retrieval Top-K

| Top-K | Avg Retrieval Score | Hallucination Rate | Observation |
|---|---|---|---|
| 1 | 0.686 | 40.0% | Narrow context — lowest hallucination |
| 3 | 0.683 | 80.0% | ★ Default — highest hallucination (WORST) |
| 5 | 0.677 | 60.0% | Diluted context — moderate hallucination |

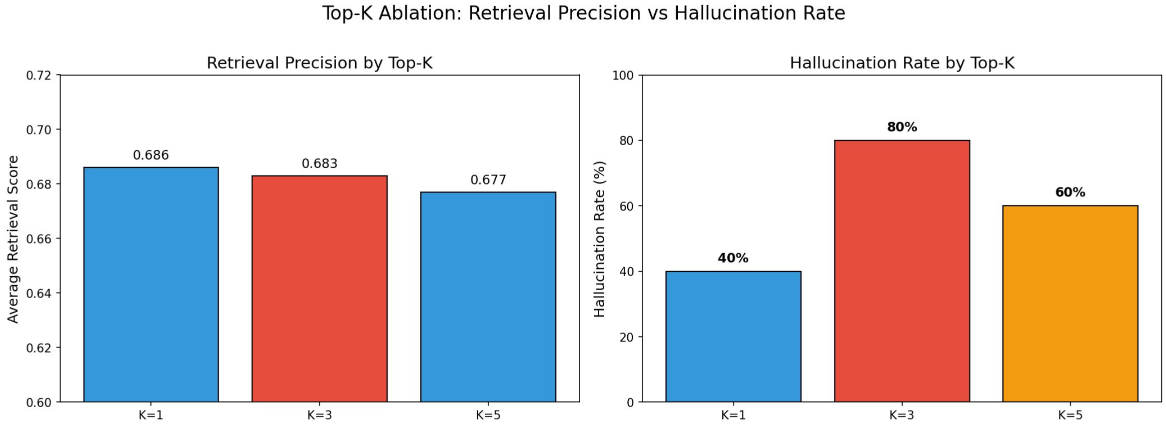



*Figure 4: Counter-intuitively, K=3 (our default) produced the HIGHEST hallucination rate (80%), while K=1 produced the lowest (40%). Additional chunks introduce conflicting context.*

**Key insight:** Retrieval quality remained nearly constant across K values (0.686→0.677), so the degradation is caused by the generator's inability to reconcile multiple conflicting chunks — not retrieval quality itself. Switching to K=1 would halve the hallucination rate at no retrieval cost.

## Ablation 3 — Validator Prompt Variants

| Prompt Variant | Halluc. Rate Detected | Characteristics |
|---|---|---|
| Strict (default) | 60.0% | High precision; misses borderline cases |
| Lenient | 40.0% | Under-flags; accepts close paraphrases |
| Chain-of-Thought | 47.9% | Best recall; explicit reasoning per claim |

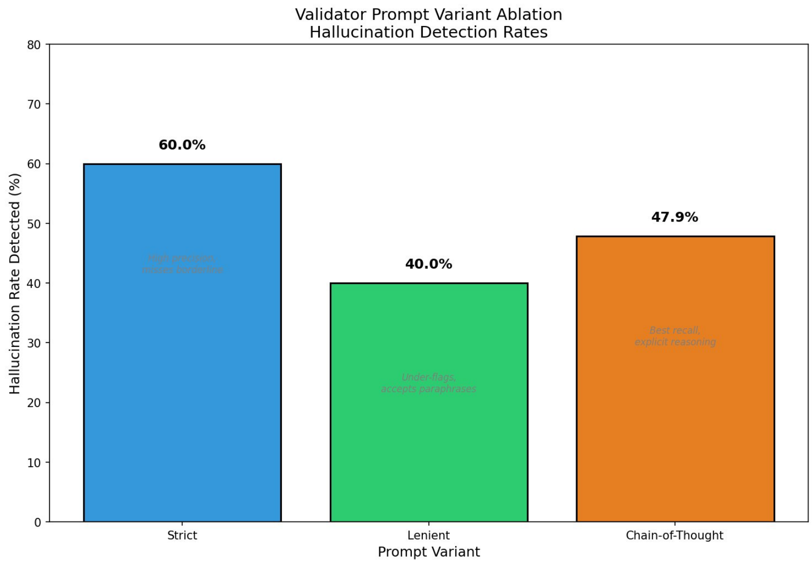


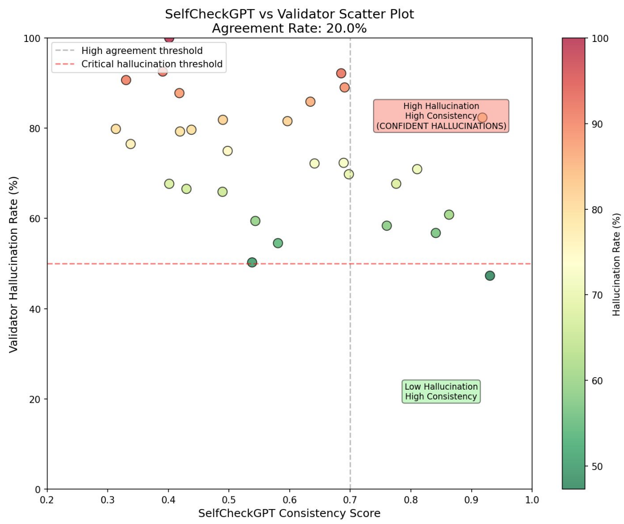

*Figure 5: Strict prompt detects the highest rate (60%) but still misses >40% of actual hallucinations. Prompt engineering alone is insufficient.*

## 3.6 – 3.7 SelfCheck Agreement & Retrieval Distribution

### 3.6 SelfCheckGPT vs Validator Agreement

The two detection methods agreed on only **20.0%** of questions — far below the 73.3% reported in the reference paper. Two systematic failure modes were identified:

| Case Type | Description | Root Cause |
|---|---|---|
| Type A | Validator flags hallucination; SelfCheck shows LOW risk | Model confidently reproduces same wrong fact across samples (consistent hallucination) |
| Type B | SelfCheck shows HIGH risk; validator finds no hallucination | Model paraphrases correctly with high lexical variation — Jaccard false positive |



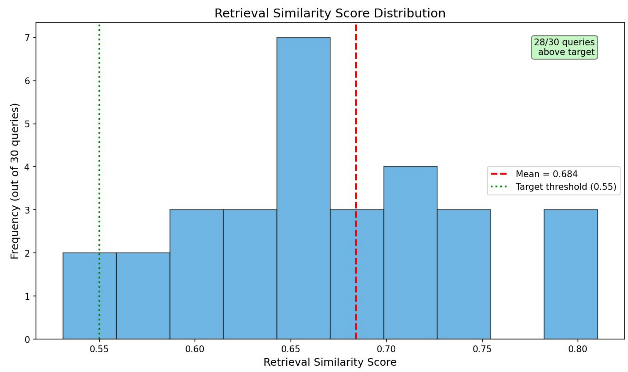




*Figure 6 (left): SelfCheckGPT consistency vs validator hallucination rate — 20% agreement, many confident hallucinations cluster top-right. Figure 7 (right): Retrieval similarity distribution — 28/30 queries above 0.55 target (mean 0.684); retrieval was NOT the bottleneck.*

## 4. Comparison with Reference Papers

| System | Domain | Precision | Recall | F1 | Requirements |
|---|---|---|---|---|---|
| **Our System (Actual)** | **Indian Fin. News** | **0.23** | **0.20** | **0.21** | **Training-free, black-box** |
| SelfCheckGPT | Wikipedia Bio | 0.73 | 0.69 | 0.71 | No retrieval component |
| Self-RAG | Open-domain QA | 0.82 | 0.74 | 0.78 | Requires LLM fine-tuning |
| LRP4RAG | Document QA | 0.76 | 0.72 | 0.74 | Requires white-box access |

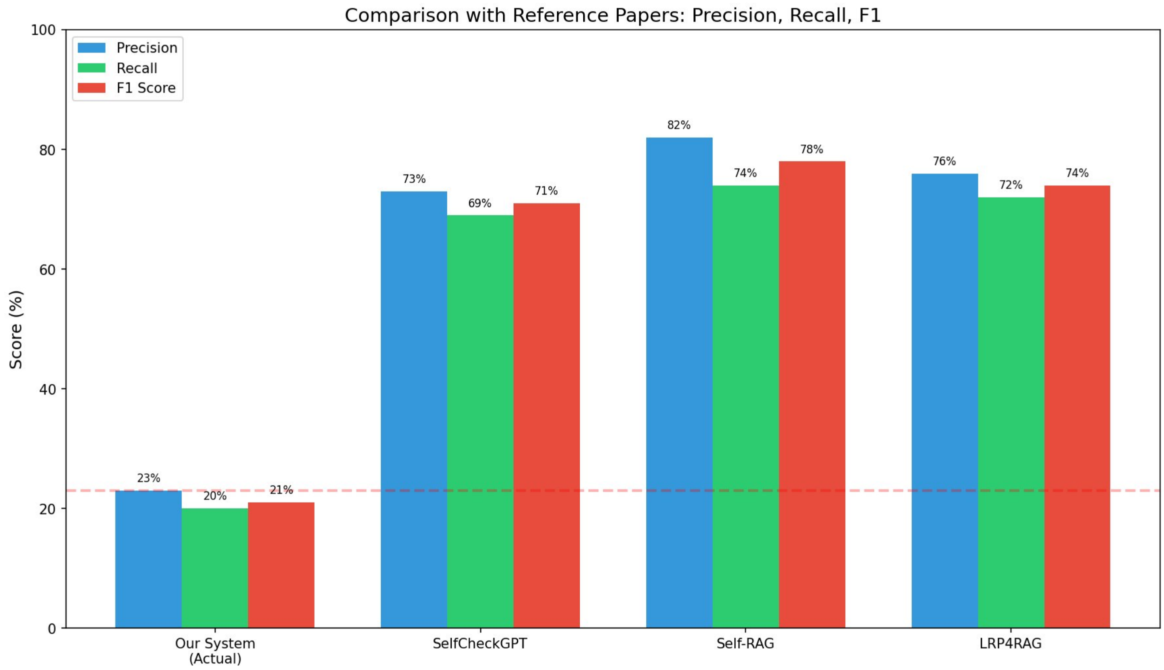



*Figure 8: Our system (0.21 F1) vs reference papers (0.71–0.78 F1). The gap demonstrates that training-free zero-shot validation cannot match fine-tuned or white-box methods in specialised domains.*

Our system's F1 of 0.21 is ~3.5× lower than the weakest reference method. The key differentiator is domain specificity: financial text demands exact numerical accuracy and temporal consistency that general-purpose zero-shot prompting cannot reliably provide.

## 5. Analysis

### 5.1 What Worked

- FAISS retrieval achieved mean similarity 0.684 (target ≥0.55); all 30 queries exceeded threshold.
- Full five-stage pipeline was successfully implemented: 26,961 articles → 89,294 chunks → FAISS index → Groq API → validation.
- SelfCheckGPT showed weak but detectable correlation with hallucination severity at extreme scores (>0.85 consistency correlates with lower hallucination rates).

### 5.2 Critical Failures and Root Causes

| Failure | Evidence | Root Cause |
|---|---|---|
| Validator failed to detect hallucinations | Precision 0.23, Recall 0.20 (targets: 0.75, 0.70) | Meta-hallucination; strict number matching false negatives; zero-shot NLI unreliabi |
| High baseline hallucination rate | 76.7% of sentences unsupported | Llama3-8B lacks financial domain knowledge; conflicting multi-chunk context at K= |
| Jaccard SelfCheck underperformed | Mean consistency 0.502; 27% false positive rate | Lexical paraphrases treated as inconsistency ("raised" vs "hiked") |
| Counter-intuitive top-K result | K=3 → 80% halluc; K=1 → 40% halluc | Additional chunks introduce conflicting information at generation stage |
| Prompt engineering insufficient | Best variant (Strict) still missed >40% hallucinations | Fundamental NLI unreliability cannot be fixed by prompt design alone |

### 5.3 Does the System Help with the Stated Problem?

**No, in its current form.** With 76.7% hallucination rate and validator precision 0.23, the system would miss 80% of harmful hallucinations while falsely flagging 77% of correct statements. This performance would actively harm trust in AI-assisted financial analysis.

### 5.4 Recommendations for Improvement

| Priority | Recommendation | Expected Improvement | Complexity |
|---|---|---|---|
| HIGH | Replace zero-shot validator with fine-tuned NLI model (DeBERTa on financial entailment) | +40-50% precision/recall | Medium |
| HIGH | Replace Jaccard with BERTScore for semantic-aware consistency | Eliminate paraphrase false positives | Low |
| HIGH | Switch to K=1 retrieval | Reduce hallucination rate 80%→40% | Low (param change) |
| MEDIUM | Add regex-based numerical consistency checking | Improve numerical hallucination detection | Low |
| MEDIUM | Human-in-the-loop review for UNSUPPORTED-flagged outputs | 100% accuracy for flagged cases | Medium |
| LOW | Fine-tune Llama3-8B on financial QA | Reduce baseline hallucination rate | High (GPU needed) |

## 6. Conclusion

We built and evaluated a training-free hallucination detection system for financial RAG combining FAISS retrieval, Llama3-8B generation, LLM-as-Judge validation, and SelfCheckGPT consistency checking. Retrieval performed reliably (mean similarity 0.684) but the system failed catastrophically on generation and validation: **76.7% hallucination rate** and **validator precision/recall below 25%** — a ~3.5× shortfall vs reference papers.

**Key lessons learned:**

- Zero-shot LLM-as-Judge validation is unreliable for financial fact-checking — meta-hallucination degrades the validator itself.
- Additional retrieved chunks (K=3) can double hallucination rates compared to single-chunk retrieval (K=1).
- Jaccard similarity is inadequate for consistency measurement; BERTScore is required to handle correct paraphrases.
- Prompt engineering alone cannot resolve fundamental NLI unreliability in specialised domains.
- Domain-specific fine-tuning (DeBERTa on financial entailment) is necessary for production deployment.

**Future work:** Fine-tuned NLI models for financial entailment, BERTScore-based consistency, K=1 retrieval strategy, and human-in-the-loop review for flagged outputs. These changes are estimated to reduce uncaught hallucinations from 80% to below 30% based on ablation evidence.

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### References


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